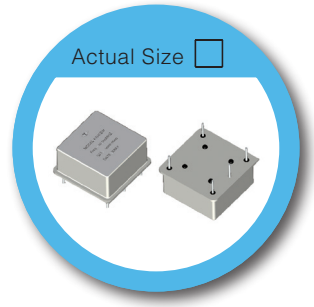
100MHz Ultra Low Noise/ Low G-Sensitivity OCXO NA-100M-6900 series



RoHS Compliant

FEATURE

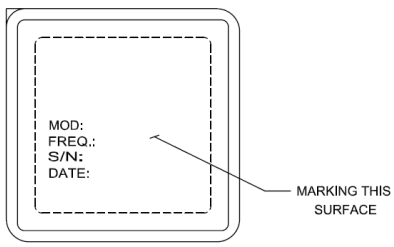
- Ultra Low Phase Noise & Low G-Sensitivity
- Hermetically Sealed Package
- Tight Frequency Stability
- Low Power Consumption
- Fast Warm-up Time
- Electrical Frequency Tuning Input
- Reference Voltage Output
- RoHS-Compliant (lead-free)

TYPICAL APPLICATION

- Instrument Reference
- Microwave Communication
- Clock Reference for Microwave Signal Source
- Test & Measurement
- Telecom Systems
- Radar Systems
- Medical (MRT)

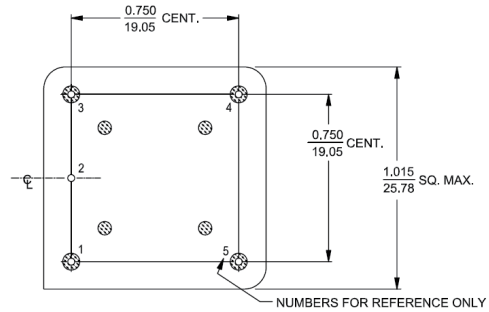
DIMENSION (mm)

[TOP VIEW]



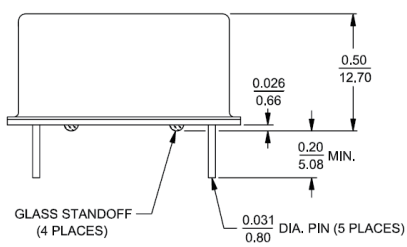
(VIEW FROM TOP)

[BOTTOM VIEW]



(VIEW FROM BOTTOM)

[SIDE VIEW]



PIN CONNECTIONS	
PIN	FUNCTION
1	R. F. OUTPUT
2	0 VOLTS & CASE
3	VCO INPUT
4	NOT CONNECTED
5	+VDC

ELECTRICAL SPECIFICATION

Test conditions: VDC = +5V; VCO = +5V; at +25 ± 3°C unless otherwise identified

OUTPUT (PIN = "R.F. OUTPUT")

Parameter	Min.	Typ.	Max.	Unit	Test Condition
Frequency (Fo)		100.000000		MHz	
Initial Accuracy	-0.3		+0.3	ppm	@ +25 ±1°C after turn on power 30 minutes Vco=+5V
Waveform		Sine wave			
Level	+15			dBm	
Load		50		Ω	+/-10%
Harmonics			-30	dBc	
Spurious			-100	dBc	

FREQUENCY STABILITY

Parameter	Min.	Typ.	Max.	Unit	Test Condition	
Ambient		±50, ±100 ±200, ±500		ppb	referenced to 25°C	**Refer to Table 1 : Ordering Information**
		-20ºC ~ +70ºC -40ºC ~ +85ºC		°C		
Aging					after 30 days of continuous operation	
Daily	-5		+5	ppb		
Yearly	-200		+200	ppb		
15 Years	-2		+2	ppm		
Voltage	-5		+5	ppb	±5% change	
Load	-5		+5	ppb	±10% change	
Short term			0.05	ppb	root Allan variance for τ=1 sec	
Warm-up	-50		+50	ppb	in 5 minutes @ +25 ±1°C	referenced to 1 hour
G-Sensitivity (each axis)			1	ppb/g		
Phase Noise (Max.)	**Option A**		**Option B**		**Refer to Table 1 : Ordering Information**	
	-100		-105	dBc/Hz	@ 10Hz	
	-130		-135	dBc/Hz	@ 100Hz	
	-157		-162	dBc/Hz	@ 1KHz	
	-180		-180	dBc/Hz	@ 10KHz	
	-185		-185	dBc/Hz	@ 100KHz	
	-185		-185	dBc/Hz	@ 1MHz	

ELECTRICAL FREQUENCY ADJUSTMENT (PIN = "VCO INPUT")

Parameter	Min.	Typ.	Max.	Unit	Test Condition
Tuning Range	±2.5			ppm	Referenced to frequency at nominal Center Voltage
Control Voltage	0		+10.0	V	
Slope		Positive			
Center Voltage		+5		V	
Linearity	-10		+10	%	

Specifications subject to change without notice.

INPUT POWER (PIN = "+VDC")

Parameter	Min.	Typ.	Max.	Unit	Test Condition
Voltage	+4.75	+5	+5.25	V	
Current					
Steady State			2.1	W	@ +25°C
During Warm-Up			950	mA	

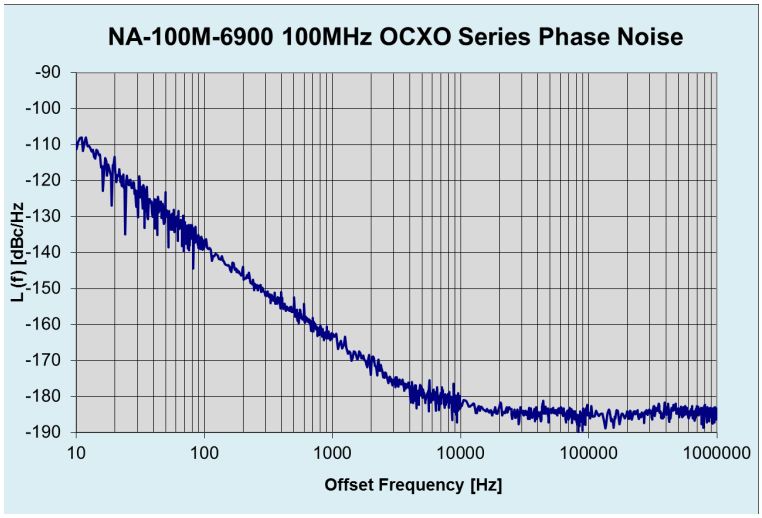
ENVIRONMENTAL

Parameter	Reference Std.	Test Condition
Operable Temperature	-40°C to +85°C	Note 1
Storage Temperature	-55°C to +105°C	
Humidity	MIL-STD-202, Method 103 Test Condition A	95% RH @ +40°C,non-condensing,240 hours
Vibration (non-operating)	MIL-STD-202, Method 201	0.06" Total p-p, 10 to 55 Hz
Shock (non-operating)	MIL-STD-202, Method 213, Test Condition J	30g, 11ms, half-sine

Note 1 : Output maintained over this temperature range. Other requirements of this specification may not be met when operating outside the temperature range in 2.1.

Table 1 : ORDERING INFORMATION

Ambient Temp. (°C) / Option		Phase Noise Option A	Phase Noise Option B
-20°C ~+70°C	±100 ppb	NA-100M-6900	NA-100M-6901
	±50 ppb	NA-100M-6910	NA-100M-6911
-40°C~+85°C	±500 ppb	NA-100M-6920	NA-100M-6921
	±200 ppb	NA-100M-6930	NA-100M-6931

Other specifications may be available upon request.

Phase Noise Test Data